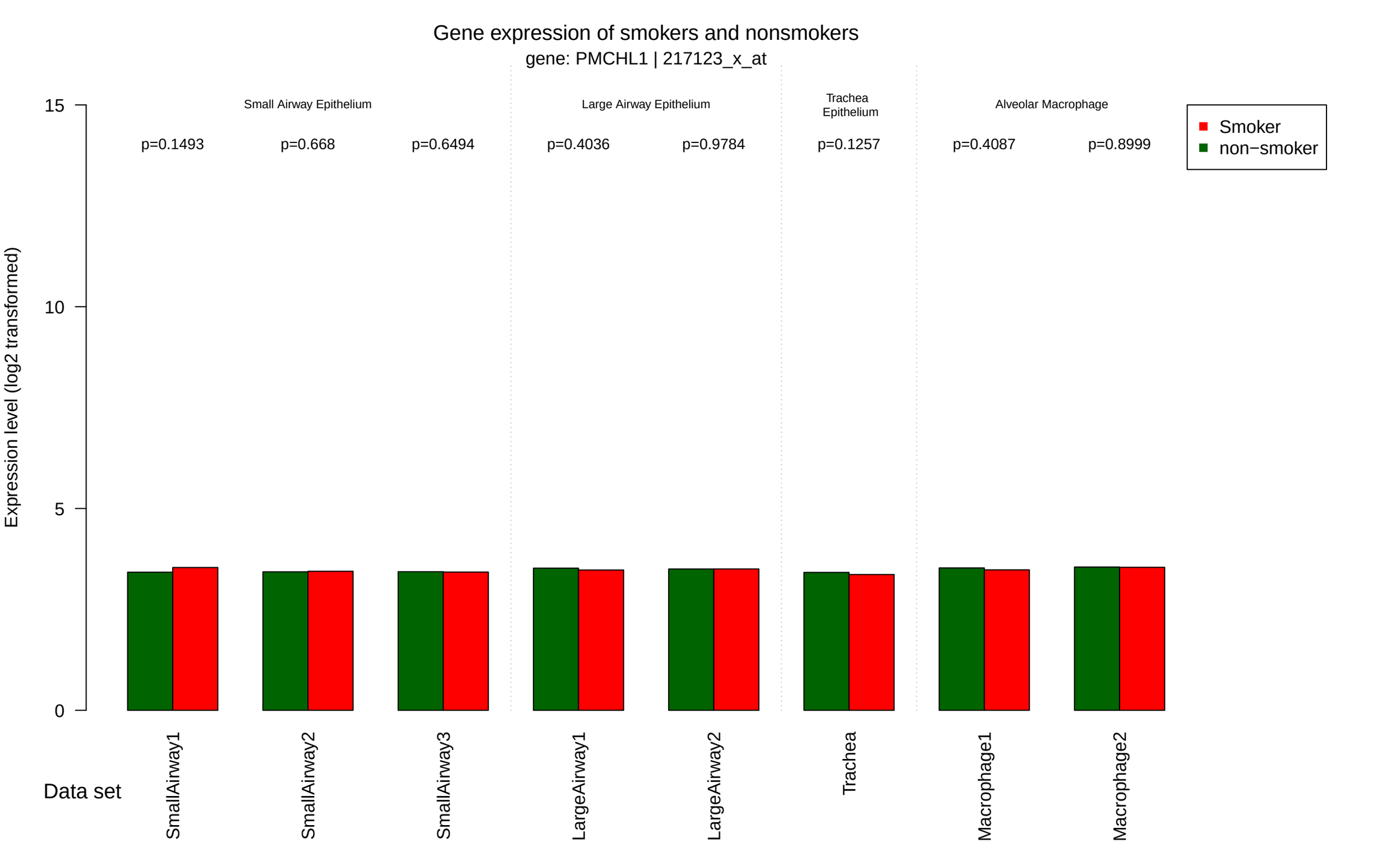
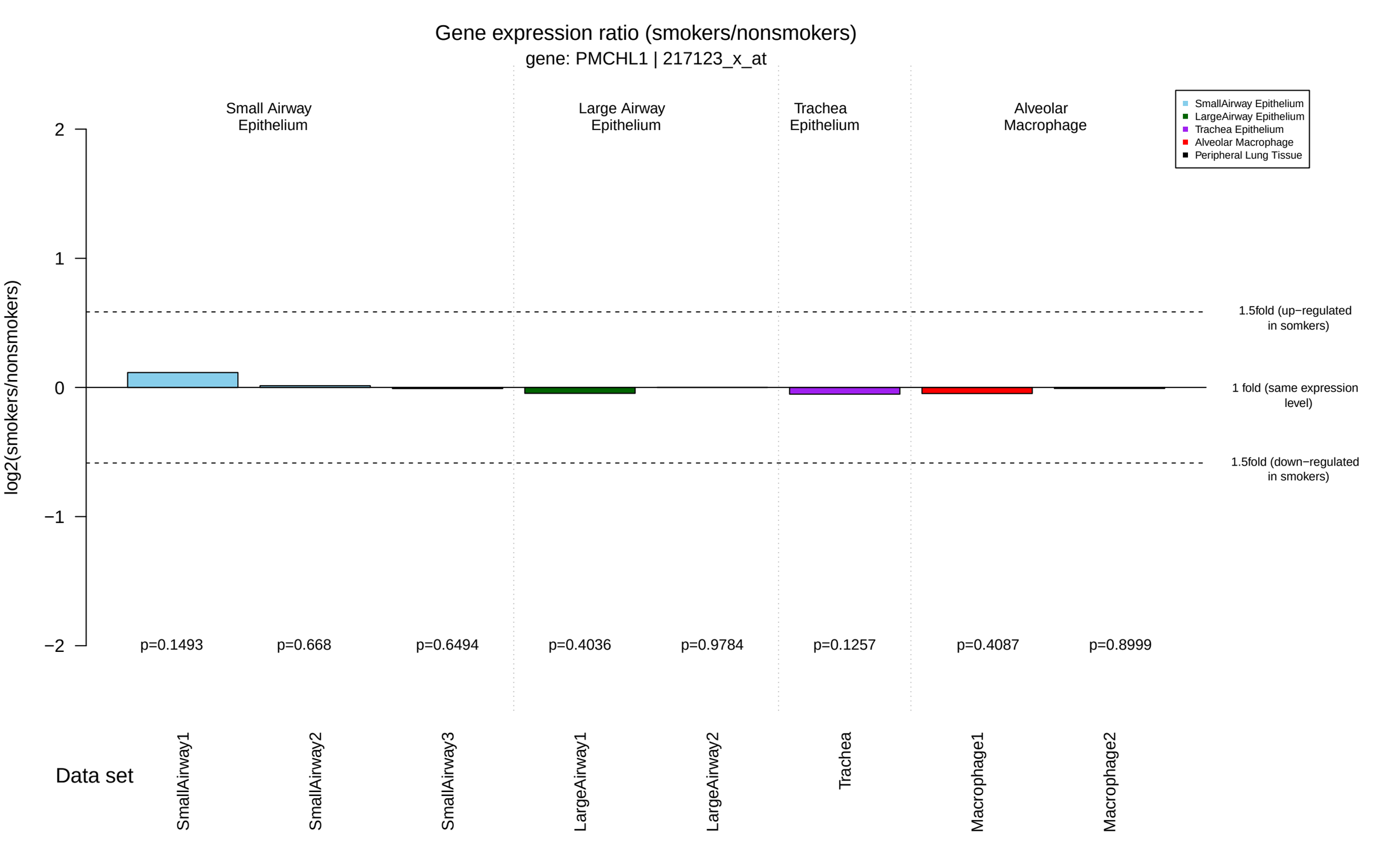
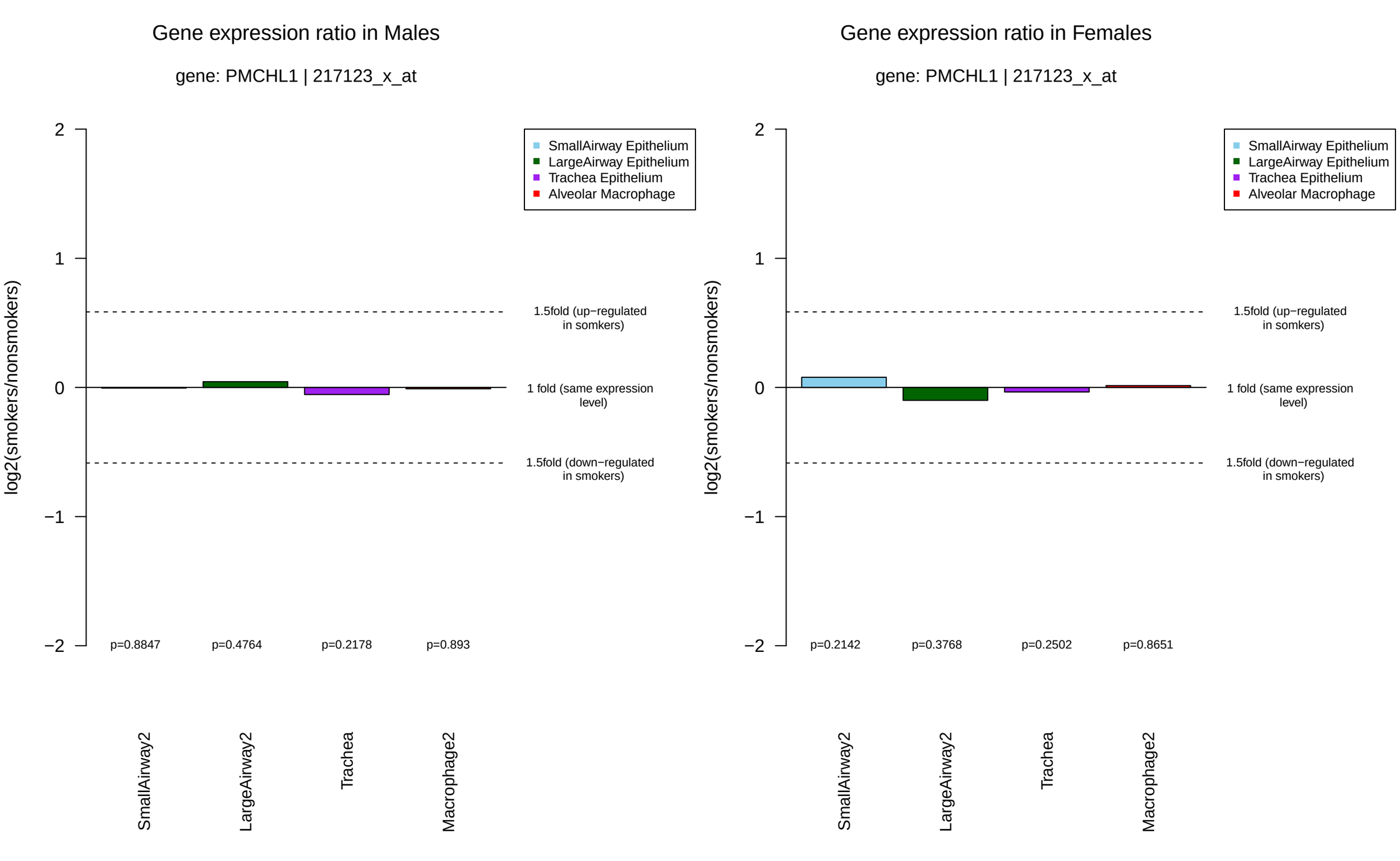
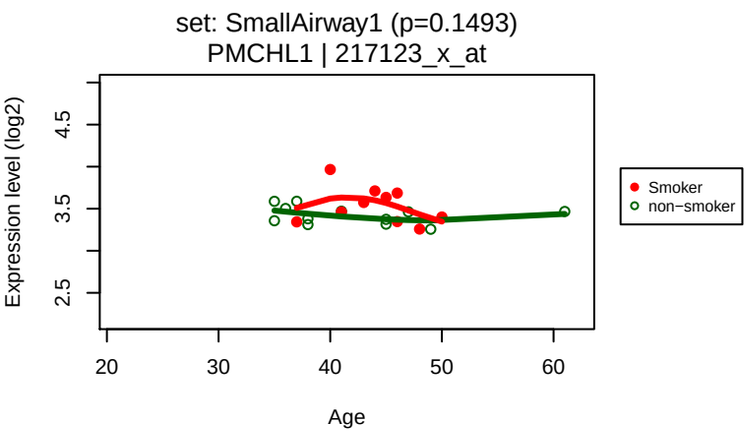
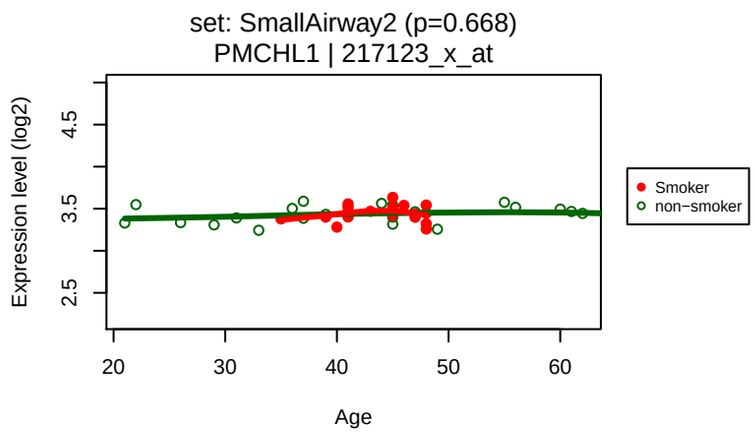
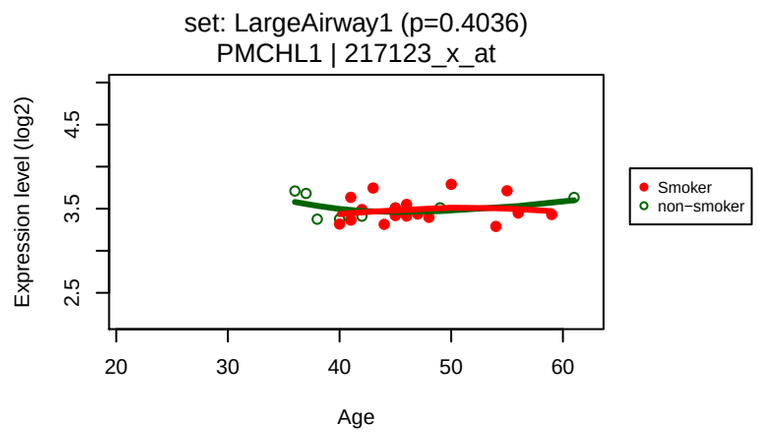

Gene expression of smokers and nonsmokers
gene: PMCHL1 | 217123_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1493
p=0.668
p=0.6494
p=0.4036
p=0.9784
p=0.1257
p=0.4087
p=0.8999
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

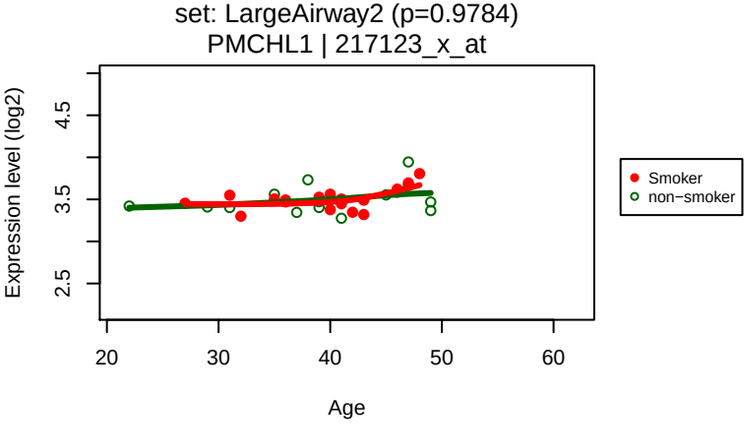

Gene expression ratio (smokers/nonsmokers)
gene: PMCHL1 | 217123_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1493
p=0.668
p=0.6494
p=0.4036
p=0.9784
p=0.1257
p=0.4087
p=0.8999
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

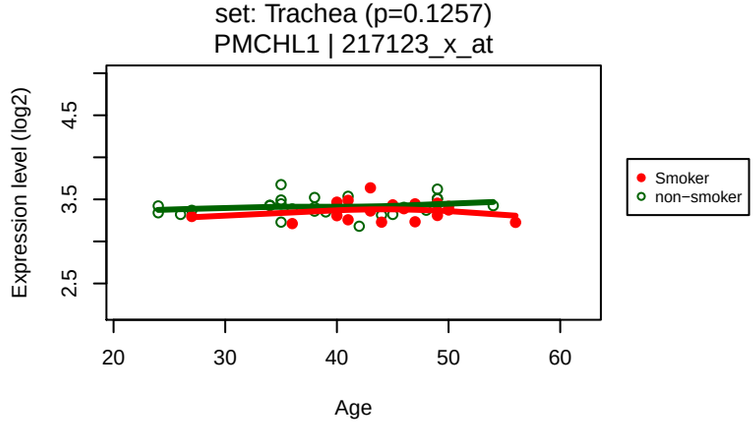

Gene expression ratio in Males
gene: PMCHL1 | 217123_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8847
p=0.4764
p=0.2178
p=0.893
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PMCHL1 | 217123_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2142
p=0.3768
p=0.2502
p=0.8651
SmallAirway2
LargeAirway2
Trachea
Macrophage2

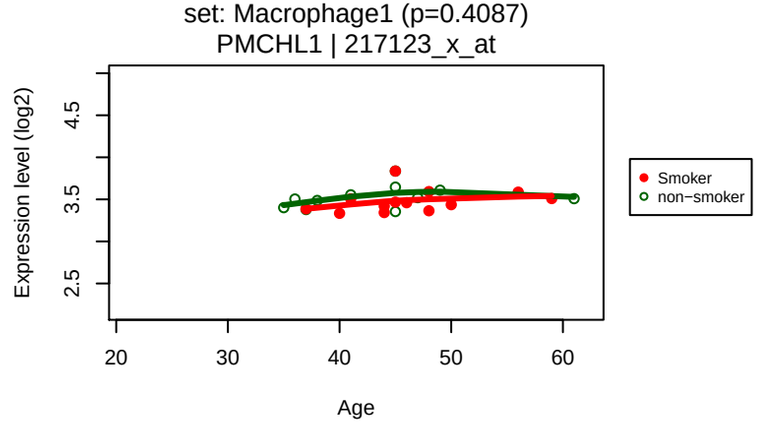

set: SmallAirway1 (p=0.1493)
PMCHL1 | 217123_x_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker

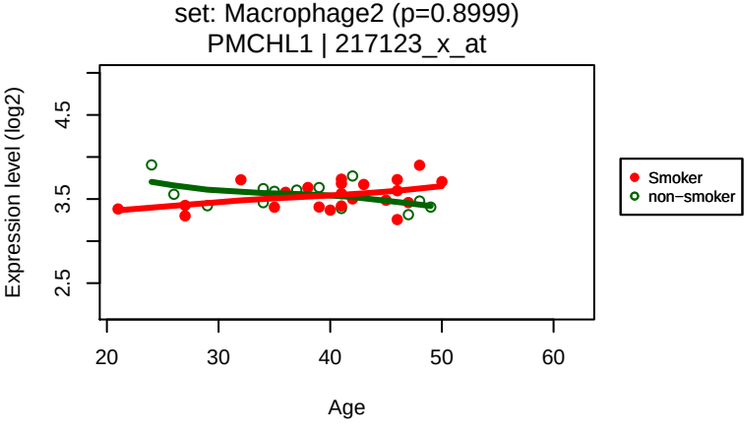

set: SmallAirway2 (p=0.668)
PMCHL1 | 217123_x_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker

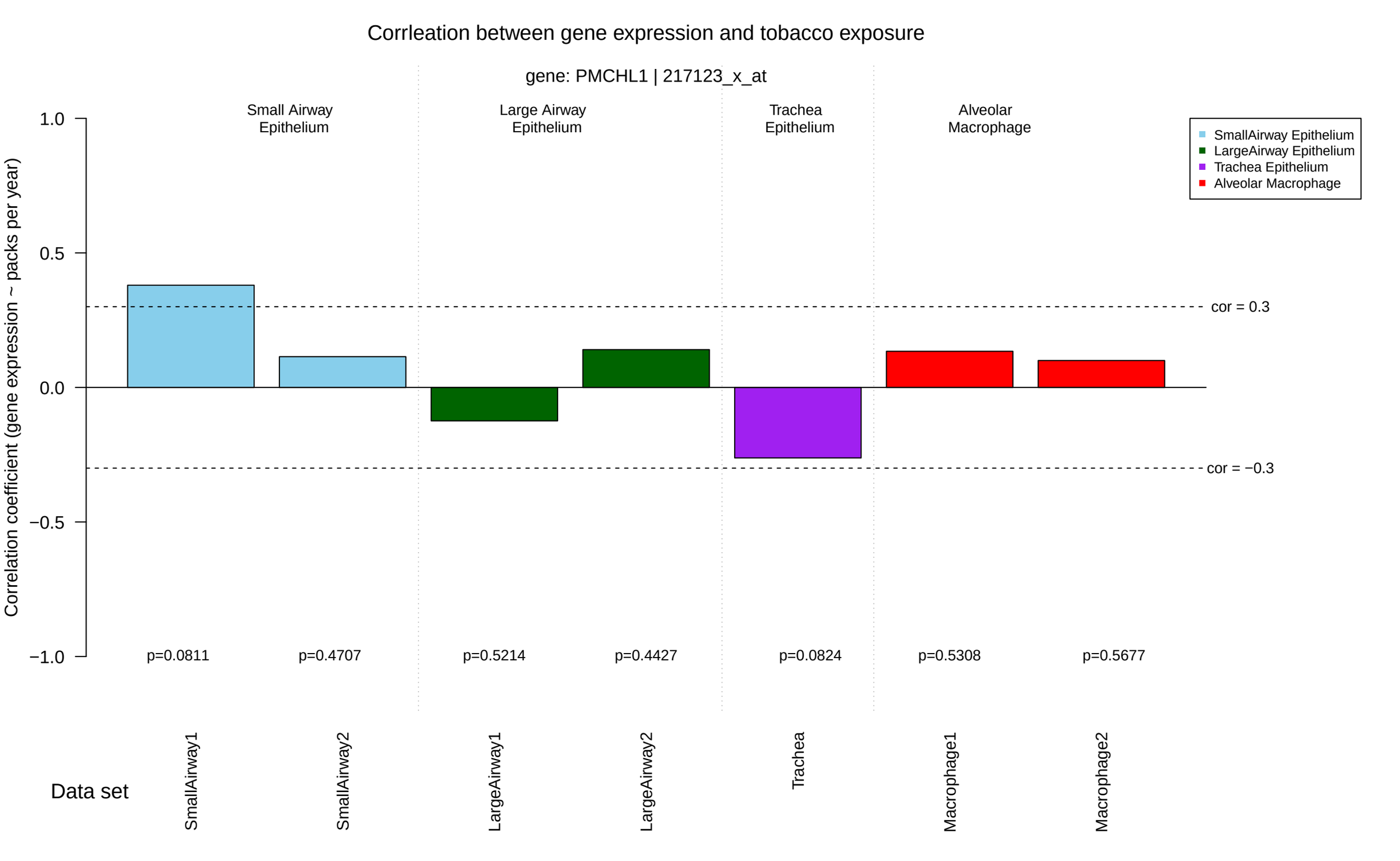

set: LargeAirway1 (p=0.4036)
PMCHL1 | 217123_x_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9784)
PMCHL1 | 217123_x_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.1257)
PMCHL1 | 217123_x_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4087)
PMCHL1 | 217123_x_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8999)
PMCHL1 | 217123_x_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PMCHL1 | 217123_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0811
p=0.4707
p=0.5214
p=0.4427
p=0.0824
p=0.5308
p=0.5677
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2